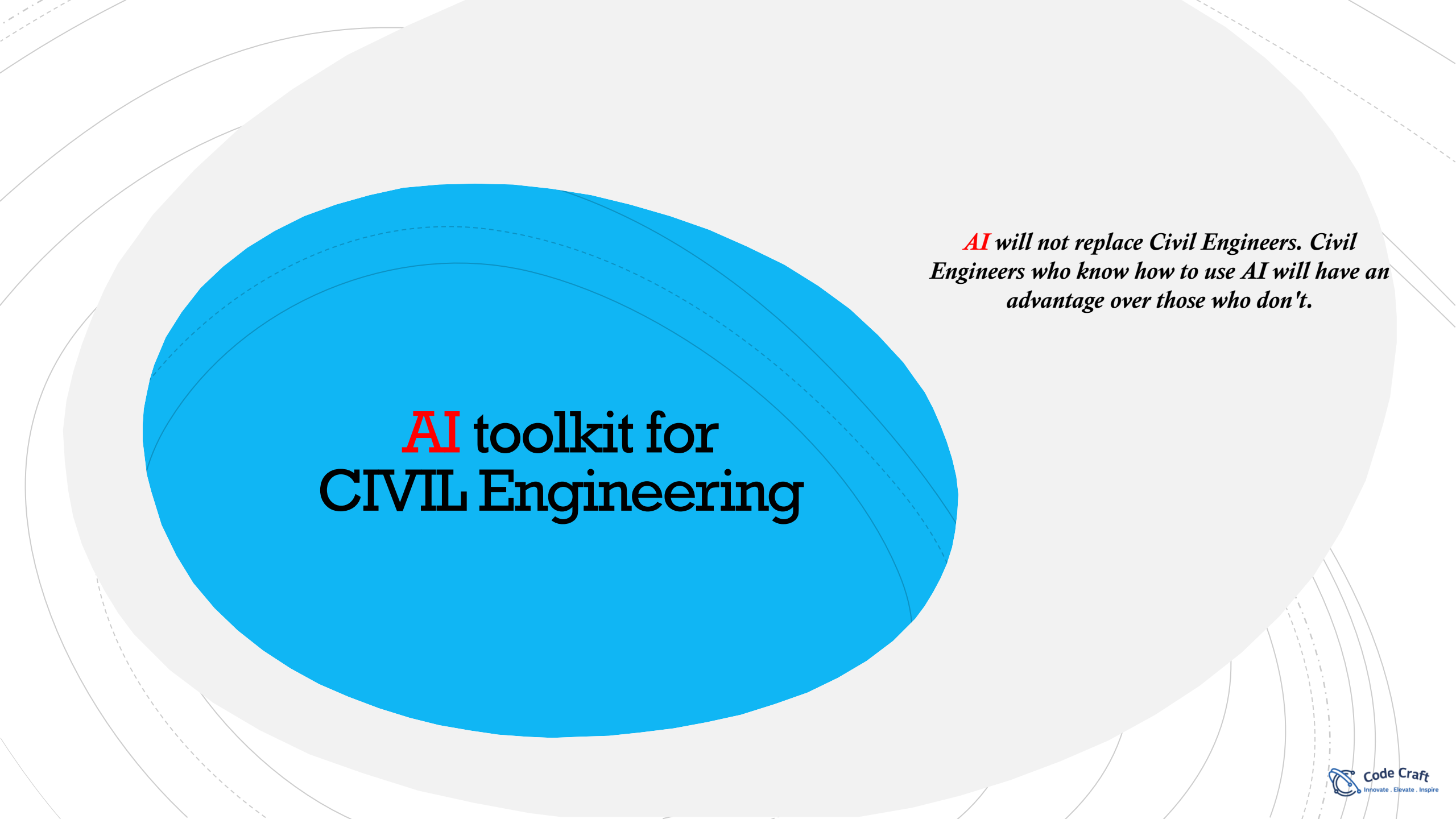


The background features a large, light gray oval shape. Inside this oval is a smaller, solid blue oval. The text "AI toolkit for CIVIL Engineering" is centered within the blue oval. The word "AI" is in red, while the rest of the text is in black. The overall design is clean and modern, with a focus on the central text.

AI toolkit for CIVIL Engineering

***AI** will not replace Civil Engineers. Civil Engineers who know how to use AI will have an advantage over those who don't.*

The lifecycle

Surveying →

Planning →

Design →

Structural Analysis →

Construction →

Safety →

Monitoring →

Maintenance

Learning Objectives

AI in Construction

Structural Design

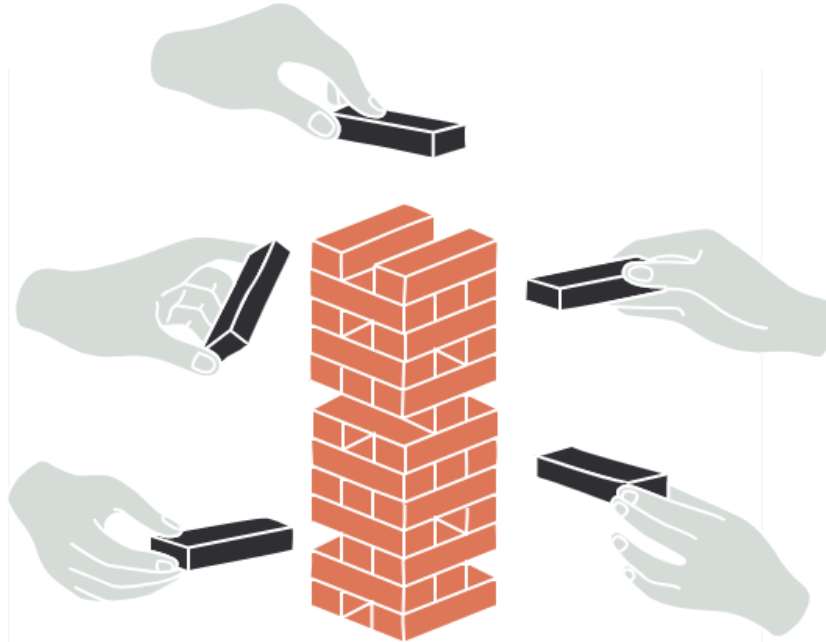
AI optimizes structural integrity

Quantity Estimation

AI automates material calculations

Surveying

AI analyzes geospatial data



BIM

AI enhances BIM workflows

Construction Planning

AI optimizes project schedules

AI Assistants

 TOOL	 WHAT ECE STUDENTS CAN USE IT FOR
 ChatGPT	 Learning concepts, debugging code, circuit explanations, project ideas, documentation
 Google Gemini	 Research, multimodal analysis, coding, technical explanations
 Claude	 Understanding large technical documents, coding and reasoning
 Microsoft Copilot	 Coding, documentation, productivity
 Perplexity	 Technical research and finding sources

AI for Structural Engineering



Tools :

[Autodesk Construction Cloud](#)
[Autodesk Forma](#)
[Autodesk Revit](#)

What you should explore

- BIM
- Generative design
- Building analysis
- Site analysis
- Construction coordination

Key concept

AI can search the design space. The engineer defines what is acceptable.

AI + BIM

Tools :
Autodesk Revit

What you should explore
BIM → AI → Intelligent Building

Intelligent Buildings. Smarter Construction.

AI-Powered BIM for Smarter Decisions Across the Project Lifecycle



AI can assist with:



Clash detection

AI identifies clashes in 3D models early before construction.



Design analysis

AI evaluates design options for structural, energy, cost and sustainability performance.



Quantity extraction

AI automates BOQ/quantity takeoff from models and drawings.



Scheduling

AI optimizes construction schedules and detects delays.



Risk identification

AI predicts project risks and suggests mitigation strategies.



Document analysis

AI reads, classifies and extracts information from contracts, RFIs, reports and drawings.



Construction coordination

AI improves communication and coordination across teams and stakeholders.



Building performance

AI monitors and predicts building performance for better operation and maintenance.



For Civil Engineering Students



Learn Smarter



Design Better



Build the Future



BIM provides the data,
AI provides the intelligence.



AI + BIM

Tools :
[Autodesk Revit](#)

What you should explore
BIM → AI → Intelligent Building



Autodesk Forma

AI-Powered Site & Design Analysis for Smarter Early-Stage Planning

Explore. Analyze. Design Better.



AI + Computational Design

Turn site data into smart design decisions.

Proposed Building Site

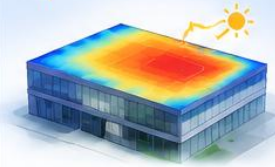


Autodesk
Forma
AI Analysis

AI-Assisted Analysis can help explore:



Solar Exposure



Find optimal orientation for energy efficiency.



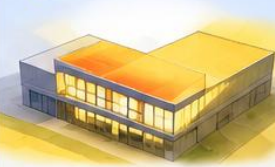
Wind



Understand wind flow and comfort conditions.



Daylight



Ensure better indoor light and comfort.



Building Orientation



Compare orientation alternatives.



Site Context



Analyze surroundings and constraints.



Massing Alternatives



Test multiple design options quickly.



Key Outcomes:

- ✓ Smarter site planning
- ✓ Better building performance
- ✓ Lower energy use
- ✓ Faster design decisions

Result:

AI-driven insights for sustainable, comfortable and cost-effective building design from the very early stages.



AI + Construction

Tools :
[Autodesk Revit](#)

What you should explore
BIM → AI → Intelligent Building



Autodesk Forma

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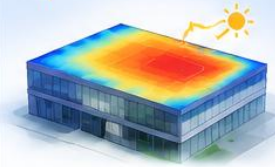


Autodesk
Forma
AI Analysis

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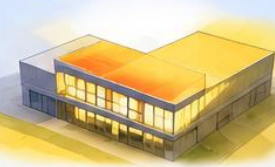
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AI + Planning & Concept Design

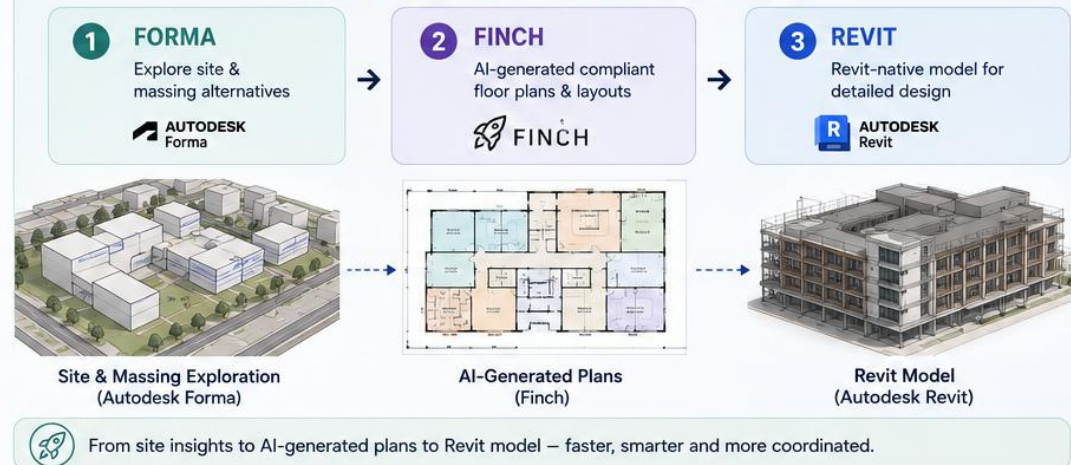
Now the question becomes:

“What should we build and how should it be arranged?”

AI can help with:

- Concept generation
- Space planning
- Alternative layouts
- Sustainability ideas
- Site utilization
- Design comparison
- Early feasibility analysis

AI-ASSISTED EARLY-STAGE PLANNING WORKFLOW



TOOLS

1 ChatGPT / Gemini

Use for: Requirements → Design alternatives

Example prompt:

Act as a civil engineering planning assistant. Propose three conceptual layouts for a G+3 academic building on a rectangular site. Consider circulation, daylight, ventilation, accessibility, parking and future expansion. Clearly state assumptions.



AI helps you think wider, evaluate faster and decide better.

2 Autodesk Forma

Forma can be used to explore early design alternatives and performance considerations before detailed design.

Autodesk is increasingly connecting Forma's planning workflows with Revit and other AEC tools.



- Sunlight Analysis
- Wind Analysis
- Daylight Analysis
- Site Metrics
- Mass Alternatives

3 Finch



Finch is an AI-assisted planning tool that turns building programs and site constraints into code-compliant floor plans in seconds.



- Code Compliant
- Intelligent Space Planning
- Multiple Alternatives
- Edit & Refine Quickly



“ Don't learn AI separately from Civil Engineering. Learn how to engineer with AI. ”



Smarter Decisions from Day One



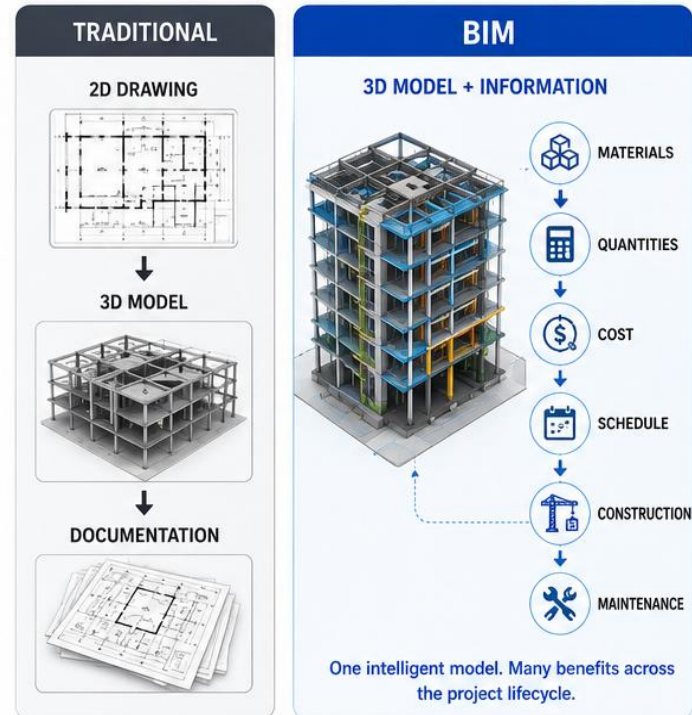
Better Designs Better Outcomes



Future Ready Engineers

AI + CAD & BIM

★ This should be one of your main demonstrations.



BIM is not just software, it's a **smarter** way to design, build and manage.

BIM – FROM DESIGN TO OPERATIONS



WHY BIM?



TOOLS

★ 1 AUTODESK REVIT

Building Information Modeling (BIM) Platform



Students should learn:

- ✓ BIM
- ✓ Architectural modelling
- ✓ Structural modelling
- ✓ Families
- ✓ Quantities
- ✓ Documentation
- ✓ Coordination
- ✓ AI Workflows

AI Revit includes AI-related workflows such as Generative Design and Autodesk Assistant.

★ 2 AUTODESK CIVIL 3D

Civil Engineering Design & Documentation



For Civil students:

- ✓ Road design
- ✓ Survey
- ✓ Terrain
- ✓ Grading
- ✓ Drainage
- ✓ Corridors
- ✓ Civil infrastructure
- ✓ AI-Assisted workflows

AI AI-assisted capabilities help automate tasks, analyze data and optimize civil design workflows.



AI TOOLS FOR CIVIL ENGINEERING

TOP APPLICATIONS & PLATFORMS WITH AI CAPABILITIES

TOOL & ICONS	WHAT IT IS GOOD AT	CIVIL ENGINEERING AREA
 AUTODESK FORMA cityplanning.com 	AI-assisted site planning, building massing, environmental analysis, wind/noise studies, and exploring design alternatives.	 Site planning / Building design
 TESTFIT testfit.com 	Quickly generates and evaluates site layouts using parameters such as units, parking, setbacks, and site constraints.	 Site planning / Feasibility
 FINCH floorplan.com 	AI-assisted generation of building layouts, floor plans, BIM geometry, and design alternatives.	 Building / BIM
 AUGMENTA engineering.com 	Spatial AI and automated building-system design; particularly useful for automated, coordinated design workflows.	 Building design / MEP
 HYPAR generative design.com 	Generative design and automation for building and engineering workflows.	 Generative design / BIM
 AUTODESK CONSTRUCTION IQ riskmanagement.com 	Uses machine learning to identify and prioritize project risks and issues.	 Construction management / Safety
 OPENSOURCE docspace.com 	AI/computer vision for site documentation, construction progress tracking, BIM comparison, and field management.	 Construction / BIM
 BUILDOTS progressmonitoring.com 	AI-based construction progress monitoring and comparison of actual site progress against the plan/BIM.	 Construction management
 DRONEDEPLOY www.dronedeploy.com 	Drone mapping, surveying, site monitoring, progress tracking, and reality capture.	 Surveying / Construction
 BRICSCAD briscad.com 	AI-assisted CAD features, drafting, modeling and design automation.	 CAD / Drafting
 AUTODESK CIVIL 3D infrastructure.com 	Civil infrastructure design: roads, terrain, grading, corridors, drainage and surveying.	 Transportation / Infrastructure
 BENTLEY OPENROADS www.bentleyopenroads.com 	Roadway design, terrain modeling, drainage and infrastructure workflows.	 Highway / Transportation
 AUTODESK REVIT www.autodeskrevit.com 	BIM modeling, structural coordination, documentation and building information management.	 Structural / BIM
 WOLFRAMALPHA wolframalpha.com 	Mathematical calculations, equations, unit conversions and engineering problem checking.	(Mathematical calculations / Engineering problem checking)
 MATLAB www.numicalanalysis.com 	Numerical analysis, data processing, modeling, simulations.	(Numerical analysis / Modeling / Simulations)

BEST TOOLS FOR STUDENTS TO LEARN FIRST

Essential AI & Engineering Tools for Civil Engineering Students

1



ChatGPT

Learn concepts, solve guided problems, generate quizzes, explain codes and learn software workflows.

2



Autodesk Forma

AI + site planning.

3



TestFit

AI + feasibility/
site layout.

4



Finch

Generative building
design.

5



Revit

BIM and structural
modeling.

6



Civil 3D

Roads, surveying
and infrastructure.

7



OPENSOURCE

OpenSpace

AI + construction
monitoring.

8



Construction IQ

AI + construction
risk management.

9



DroneDeploy

DroneDeploy

AI + surveying/
mapping.

10



MATLAB | **Wolfram|Alpha**

**MATLAB /
Wolfram|Alpha**

Engineering calculations
and data analysis.



Start Simple, Learn Deep, Build Smart.

Master these tools to enhance learning, design better, and build the future of smart and sustainable infrastructure.



THE EVOLUTION OF CIVIL ENGINEERING

🕒 YESTERDAY

Drawing + Calculation + Construction

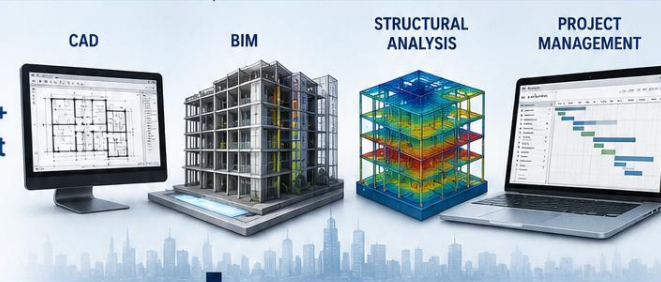
- Manual drawings
- Hand calculations
- On-site execution
- Experience driven



📅 TODAY

CAD + BIM + Structural Analysis + Project Management

- 2D/3D CAD drafting
- BIM for better coordination
- Structural analysis & simulation
- Project planning & collaboration



🚀 TOMORROW

AI + BIM + GIS + Computer Vision + IoT + Robotics

- Smarter data, better decisions
- Real-time monitoring & insights
- Automation & autonomous systems
- Predict, optimize & transform



INTELLIGENT CIVIL ENGINEER



DATA DRIVEN
DECISIONS



PREDICT &
OPTIMIZE



COLLABORATE
SMARTER



AUTOMATE &
INNOVATE



SUSTAINABLE
FUTURE



BUILD BETTER,
BUILD SAFER

“ Don't learn AI separately from Civil Engineering.
Learn how to engineer with AI. ”

Resources

https://www.wolframalpha.com/?utm_source=chatgpt.com

https://www.csiamerica.com/products/etabs?utm_source=chatgpt.com

https://www.csiamerica.com/products/sap2000?utm_source=chatgpt.com

https://www.bentley.com/software/staad/?utm_source=chatgpt.com

https://www.ansys.com/?utm_source=chatgpt.com